



DEPARTMENT OF INFORMATION TECHNOLOGY
LAB MANUAL

CS23432 – Software Construction

(REGULATION 2023)

RAJALAKSHMI ENGINEERING COLLEGE

Thandalam, Chennai-602015

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Year / Branch / Section: **2nd / B.Tech IT / 'C'**

Semester: **IV**

Academic Year: **2025 - 2026**



RAJALAKSHMI ENGINEERING
COLLEGE
An Autonomous Institution

BONAFIDE CERTIFICATE

Name: MOHAMMED DHARICA - I - B

Academic Year: 2025-2026 Semester: IV Branch: B.TECH - IT

Register No

2116241001137

Certified that this is the bonafide record of work done by the above student in
the SOFTWARE CONSTRUCTION Laboratory

during the academic year 2025 - 2026

Signature of Faculty in-charge

Submitted for the Practical Examination held on 12/05/2026

Internal Examiner

External Examiner

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I N D E X

NAME: Mohammad Dhanig J.B. B. Tech "IT" STD: 241001137 SEC: ROLL NO: SUB: Software construction

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Ex. no: 1

Azure DevOps Environment Setup

21101126

Aim:

To setup and access the Azure DevOps environment by creating an organization through the Azure portal

Installation:

1. Open your web browser and go to the azure website: <https://azure.microsoft.com/en-us/get-started/azure-portal>. Sign in using your Microsoft account credentials.

If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>

Get started with azure

Sign in

2. Azure home page

Azure services - Azure DevOps organizations

3. Open DevOps environment in the Azure platform by typing ~~Azure~~ DevOps organization in the search bar.

4. Click on the my azure DevOps organization link and create an organization and you should be taken to the Azure DevOps organization Home page.

Azure DevOps.

Result:

Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal

Ex. NO: 2

Azure DevOps project setup and
User story Management.

Aim:

To setup an Azure DevOps project for efficient collaboration and agile work management.

Procedure:

1. create an Azure account,

Azure DevOps

mohammeddhanir .ib. 2024. it@ rufalakshmi.edu. in

Name your Azure DevOps organization

dev.azure.com/ mohammeddhanir .ib. 2024. it

2) Create the first project in your organization

(a) After the organization is setup, you'll need to create your first project. This is where you'll begin to manage code, pipelines, work items and more.

(b) On the organization home page, click on the New project button.

(c) Enter the project name, description and visibility options

Name: choose a name for the project

Description: optionally, add a description to provide more context about the project

Visibility: Choose whether you want the project to be private (accessible only to those invited) or public (accessible to anyone)

(d) Once you've filled out the details, click create to setup your first project.

Create New Project

Project Name: Social Network Platform

visibility: Public

Version Control: Git

Work item process: Mohammed Dharig, ib. 2024, it Agile

Create

3. Once logged in, ensure you are in the correct organization or you're part of multiple organizations, you can switch between them from the top left. Corner (next to your user profile). Click on the organization name, you should be taken to the Azure DevOps Organization Home page.

MD

Mohammed Dharig

mohammed.dharig.ib.2024.it@rajalakshmi.edu.in

dev.azure.com / mohammeddharig.ib.2024.it (owner)

Projects - Social Network Platform

4. ~~Project~~ dashboard.

About this Project:

The Social Network Platform is a user-friendly web application that allows users to create profiles, connect with friends, and share posts, photos and videos. It includes features like friend requests, messaging and notification for better communication. The platform ensures safety through content moderation and basic privacy controls, allowing users to manage visibility and block or unfriend others.

5. To manage user stories:

- (a) From the left-hand navigation menu, click on Boards. This will take you to the main Boards page, where you can manage work items, backlogs and sprints.
- (b) On the work items page, you'll see the option to Add a work item at the top. Alternatively, you can find a + button or Add New work Item depending on the view you're in. From the Add a work item dropdown, select User story. This will open a form to enter details for the new user story.

Boards → work items → + New work item → user story

Result:

Successfully created an azure DevOps project with user story management and agile workflow setup.

Ex. No-3

Setting up epics, Features and user stories
for project planning

11/02/26

Aim:

To learn about how to create epics, user story, features, backlogs for your assigned project.

Procedure:

create Epic, Features, User stories, Task:

As a ~~user~~, I want to connect with people, share content, and communicate safely so that I can build relationships and interact effectively online

1. Fill in Epics

Epic 1

User management and profile system

2. Fill in Features

1- User registration and Login

3- user Profile Management

3. ~~Fill~~ in user story Details.

4- Registration and Log In

5- creation and updation of profile

6- Privacy options

4. As a ~~user~~, I want to register a new account and log in securely so that I can access and use the social networking platform.

5. As a user, I want to create my profile and update my personal information so that my account details stay accurate and visible according to my preferences.

6. As a user, I want to control my privacy settings so that I can decide who can view my profile, posts, and personal information.

Result:

Thus, the creation of epic, features, user story and task has been created successfully.

Ex. NO. 11

Sprint Planning

25/10/2026

Aim:

to assign user story to specific sprint for the social network platform

Sprint Planning

Sprint 1:

Taskboard

1. core Social Feature Development

Assigned to: Mohith Roshan T

Iteration 1:

8. user registration and login system

Assigned to: Mukesh Raja V

Iteration 2:

9. Send / accept / reject friend requests

assigned to: Mukesh Raja V

10. Implement private messaging

assigned to: Mohith Roshan T

Iteration 3:

11. Post text, photos and videos

Assigned to: Madhavan R

Result:

The sprints are created for the social network platform.

Ex. No. 5

Poker Estimation

11103126

Aim:

To create poker estimation for the user stories -
Social Network Platform.

Poker Estimation:

Story points measure:

- Effort
- Time
- Complexity

This uses Fibonacci series: 1, 2, 3, 5, 8, 13, ...

Story points for Social Network platform

User story	Story points	Reason
Registration and Login	5	Core Functionality with validation and authentication setup
Creation and updation of Profile	3	Simple CRUD operations with moderate UI handling
Privacy Options	8	Complex logic for access control and user-specific settings

Total $\Rightarrow$ 16 points (for Epic 1)

Breaks work items $\rightarrow$ User story $\rightarrow$ story points $\rightarrow$ (1/2/3/5) $\rightarrow$ Save

repeat for all user stories

Result:

The Estimation/ Story points is created for the project Social network platform using poker estimation.

Ex. no: 6 Designing class and sequence

Date: 18/03/26 Diagrams for Project Architecture

Aim:

To design a class diagram and Sequence diagram for the Social Network Platform project

Class Diagram:

It represents the static structure of the social network platform, showing the system's classes, their attributes, methods, and the relationships between them.

Key classes:

- * User - userId, name, email, password, hash, bio, profile, privacy
- * Post - postId, content, mediaUrl, visibility, likes, comments
- * Friendrequest - requestId, senderId, receiverId, status, sentAt
- * Message - messageId, senderId, receiverId, content, timestamp
- * Notification - notificationId, userId, type, message, isRead
- * ContentModerationLog - logId, postId, flagType

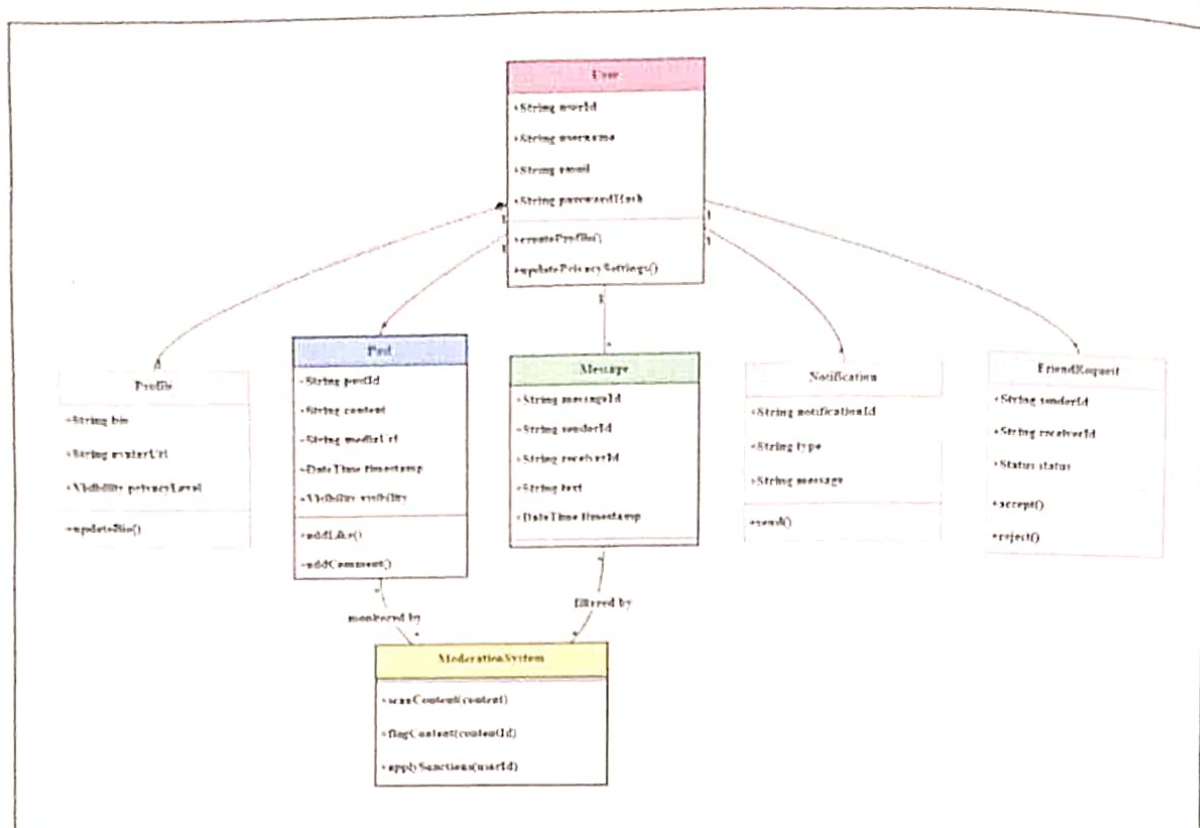
Sequence diagram:

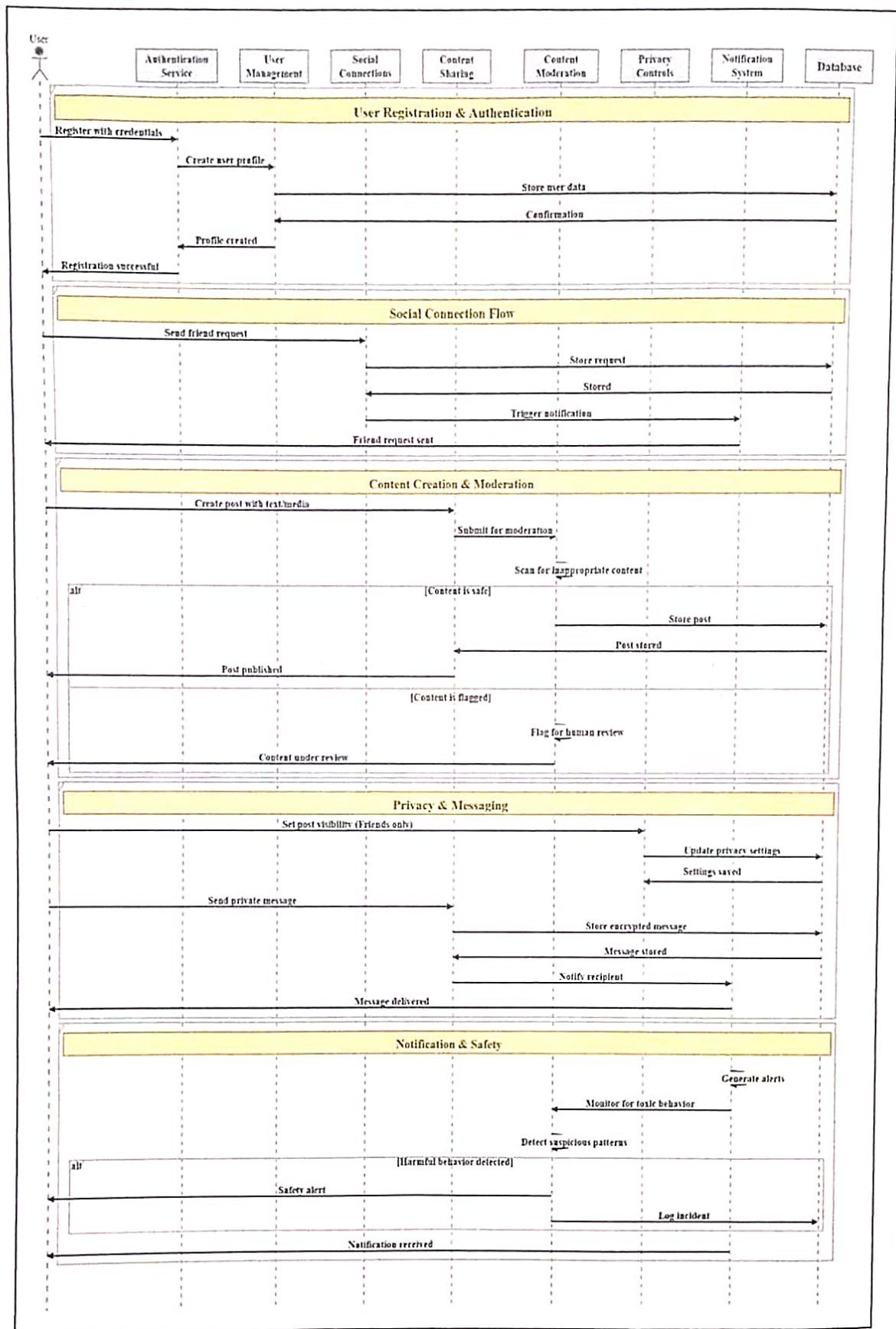
It shows the dynamic interactions between objects for the key use case: User Login and Post creation with Content moderation.

Sequence Flow - User Login:



P.T.O.





User enters credentials → AuthController validates → UserDB checks
 → session created → dashboard loaded

Sequence Flow - create post with moderation:

User submits post → Post Controller receives → CMS scans
 If clean: PostDB saves, Feed updated → Notification sent to followers.

If flagged: Post held for review → Admin notified → Admin reviews → Approve or reject.

Result:

The class Diagram and sequence diagram is designed successfully for the social network platform.

Ex.no: 7 Designing architectural and ER diagrams
Date: 25/03/26 for project Structure

Aim:

To design an architectural diagram and ER diagram for the social network platform Project

Architectural Diagram:-

The diagram shows the high-level system design of the social network platform using a layered cloud-based architecture.

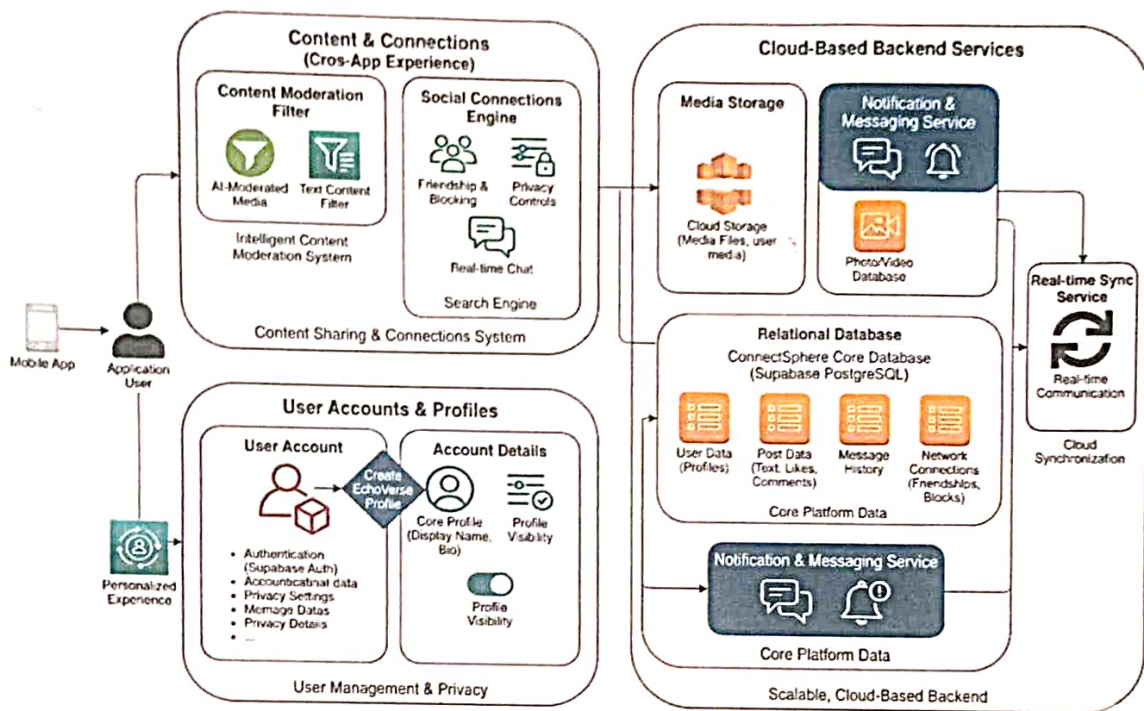
Architecture Layers;

- * Presentation Layer - React.js / Mobile App Frontend
- * API Gateway Layer - Load balancer, request routing, authentication middleware
- * Service Layer - Microservices for:
User service, Post service, Messaging service, notification service, Content moderation, admin Service.
- * Data Layer - PostgreSQL, Redis, cloud storage
- * Monitoring & Logging - Azure monitor, Application insights.

ER diagram:

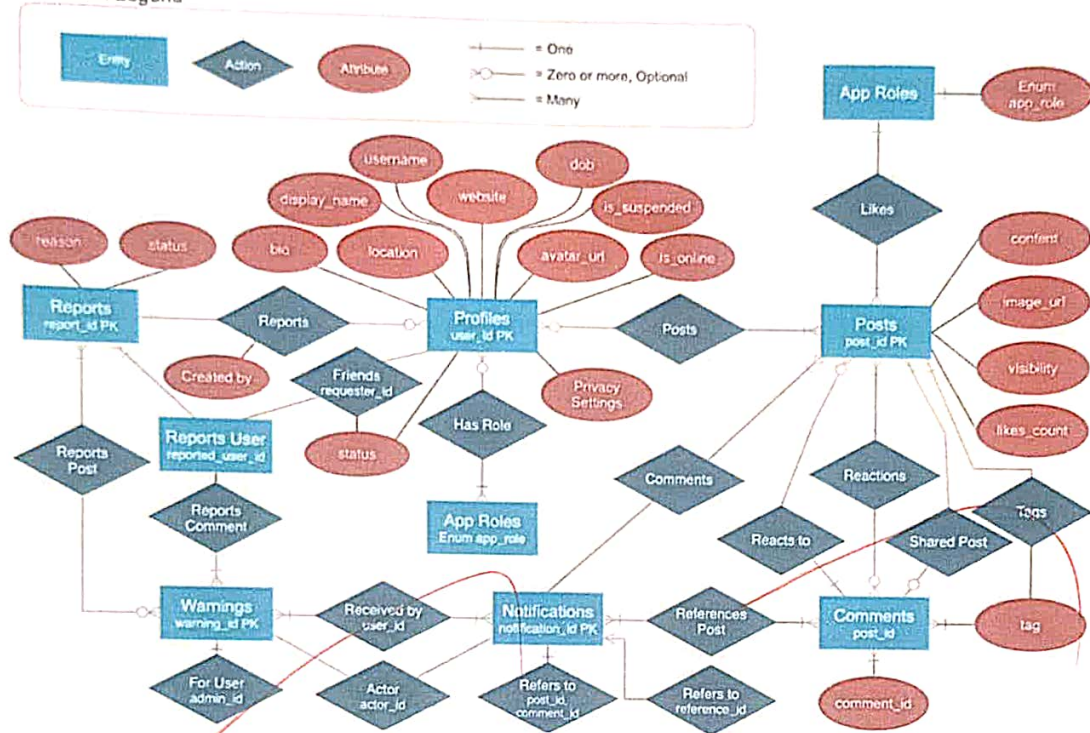
It represents the database schema showing all entities and their relationships in the platform.

EchoVerse Social Networking Platform - Simplified Architecture Diagram



EchoVerse Software Architecture Diagram

Notation Legend



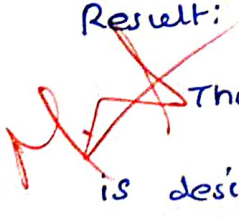
Key entities & relationships:

- USER (Id, name, email, password, bio, profile, createdAt)
- POST (Id, content, mediaurl, visibility, createdAt)
- FRIEND_REQUEST (requestId, senderId, receiverId, createdAt)
- COMMENT (id, postId, content, createdAt)
- LIKE (likeId, postId, userId, likedAt)
- NOTIFICATION (notifyId, userId, type, message, isRead, createdAt)
- CONTENT_FLAG (flagId, postId, flaggedBy, reason, reviewedAt)

Relationships:

- USER || -- o { POST : creates
- USER || -- o { FRIEND request: sends/receives
- POST || -- o { COMMENT : has
- POST || -- o { LIKE: receives
- USER || -- o { MESSAGE: sends/receives
- POST || -- o { CONTENT_FLAG: get Flagged

Result:

 The Architecture diagram and ER Diagram is designed successfully for the Social Network Platform.

Ex. no: 8

Testing - Test Plans and Test cases

Date: 01/04/26

Aim:

Test Plans and Test Cases and write two test cases for ~~at~~ least five user stories showcasing the happy path and error scenarios in Azure DevOps platform.

Test case Design Procedure

1. Understand Core Features of the Application

- User Registration & Login
- profile creation and management
- Friend request management
- Post creation and News Feed
- content moderation & safety

2. Define user interactions:

- Each test case simulates a real user behaviour

3. Design Happy path Test cases

- Focused on validating that all features function as expected under normal conditions.
- Example: User registers successfully, creates a post, or sends a friend request.

4. Design Error Path Test Cases:

- simulate negative or unexpected scenarios to test robustness and error handling.

Handwritten notes at the top of the page:

- 1. The first step is to create a test plan.
- 2. The second step is to create test cases.
- 3. The third step is to create test suites.
- 4. The fourth step is to create test runs.
- 5. The fifth step is to create test artifacts.

Screenshot of the Azure DevOps interface showing the 'Test Plans' section. The left sidebar lists various options: Overview, Boards, Tests, Pipelines, Test Plans, Test plans, Progress report, Parameters, Configurations, Runs, Exploratory sessions, Artifacts, and Project settings. The 'Test Plans' section is selected, showing a list of test plans under 'Social Network Platform - Test Plan'. The 'TS01 - User Registration & Login' test plan is highlighted. The main pane displays the 'TS01 - User Registration & Login (ID: 63)' test plan details, including a table of test cases.

Order	Test Case ID	Assigned To	State
1	64	Mohammed Ali	Done
2	72	Mubashir Raja	Done

TEST CASE 64

64 TC01 - Successful User Registration

Mohammed Othman I B

0 Comments Add Tag

Save and Close Follow

State Design Area Social Network Platform

Reason Type Regression Social Network Platform/Iteration 1

Updated by Mohammed Othman I B Wednesday

Steps Summary Associated Automation

Recent test results

- Passed by with Windows 10 Completed 7:01 AM 4/22/2026
- Not Applicable by with Windows 10 Completed 7:01 AM 4/22/2026
- Passed by with Windows 10 Completed 7:01 AM 4/22/2026

[View all test result history](#)

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Steps

Step Action Expected result

- Go to the Sign-Up page Registration form is displayed.
- Enter valid name, email, and password Fields accept values without error.
- Click "Register" Account is created and user is redirected to the dashboard.

Click or type here to add a step

Parameter values

TEST CASE 72

72 TC02 - Login with Invalid Credentials

Mukesh Raja V

0 Comments Add Tag

Save and Close Follow

State Design Area Social Network Platform

Reason Type Regression Social Network Platform/Iteration 1

Updated by Mukesh Raja V Wednesday

Steps Summary Associated Automation

Recent test results

- Run manual tests with web runner. [Learn how to get started](#)

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link to Azure DevOps commit will automatically branch to

Steps

Step Action Expected result

- Go to the Login page. Input is accepted.
- Enter a valid email with an incorrect password. Error message "Invalid username or password" is shown.
- Click "Login" User is NOT redirected to the dashboard.

Click or type here to add a step

Parameter values

5. Break Down steps and Expected results:

- Each test case contains step-by-step actions and a corresponding expected outcome.
- Ensures clarity for both testers and automation scripts

6. Use clear naming and IDs:

- Test cases are named clearly
- Helps in quick identification and linking to user stories or features.

7. Separate test suites:

- Grouped test cases based on functionality
- Improves organisation and test execution flow in Azure DevOps.

8. Prioritise and Review:

- critical user actions are marked high-priority
- Reviewed for completeness and traceability against feature requirements.

Test Plans

User stories:

- As a user, I want to register and log in securely so that I can access my account
- As a user, I want to create and update my profile so that I can express my identity.

Social Network Platform

Apr 22
Apr 25
10% only, 100% passed. View report

Test Suites

- Filter suites by name
- Social Network Platform - Test Plans
- TC01 - Control Moderation [5]
- TC02 - Post Creation and ...
- TC03 - Profile Creation and ...
- TC04 - User Registration & ...

Test Plans

- Test plans
- Progress report
- Parameters
- Configurations
- Items
- Explorative sessions
- Artifacts

Project settings

TC03 - Profile Creation and Management (ID: 68)

Define Execute Chart

Test Cases (2 items)

Test Case ID	Test Case Name	Assign to	Due
TC03	Successfully Create Profile	MADHAVAN R	Due
TC04	Save Profile with Empty Required Fields	MADHAVAN R	Due

TEST CASE 73

73 TC03 - Successfully Create Profile

MADHAVAN R

0 Comments Add Tag

Tags: Design Area: Social Network Platform

Version: None Regression: Social Network Platform/Iteration 1

Steps

- Log in with valid credentials.
- Navigate to 'Edit Profile'.
- Enter name, bio and upload a profile photo.

Click or type here to add a step

Expected result

Profile is updated and changes are reflected immediately.

Recent test results

Run manual tests with web runner. [Learn how to get started](#)

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure DevOps resource and connect the boards to

TEST CASE 77

77 TC04 - Save Profile with Empty Required Fields

MADHAVAN R

0 Comments Add Tag

Tags: Design Area: Social Network Platform

Version: None Regression: Social Network Platform/Iteration 1

Steps

- Navigate to 'Edit Profile'.
- Clear the name field and click 'Save'.

Click or type here to add a step

Expected result

Error message 'Name cannot be empty' is displayed.
Profile is NOT saved.

Recent test results

Run manual tests with web runner. [Learn how to get started](#)

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure DevOps resource and connect the boards to

Test suit

1. TCO1 - Successful user registration

- Action:

- Go to the sign-up page
- Enter valid name, email and password
- click "Register"

- Expected Results:

- Registration form is displayed
- Account is created and user is redirected to the dashboard.

Type: - Happy Path

2. TCO2 - Login with Invalid credentials

- Action:

- Go to the Login page
- Enter a valid email with an incorrect password
- Click "Login"

- Expected Results

- Input is accepted
- Error message if invalid username or password
- User is NOT redirected to the dashboard

-Type: Error Path

Result:

The test plans and test cases for the user stories are created in Azure DevOps with Happy path and Error path for the platform.

Ex-no: 9

Date: 08/04/26

Load Testing and pipelines

Aim:

To perform load testing using Azure portal Load Testing and create an Azure DevOps pipeline for automation.

Azure Load Testing:

It is used to simulate high traffic on web apps and APIs to analyse performance, identify bottlenecks, and ensure scalability. It helps verify that the social network platform can handle multiple users and services reliably.

Steps to create azure Load Testing resource:

1. Login to azure portal.
2. Search and select azure Load Testing → click create.
3. Enter details like subscription, resource group, name and location.
4. Click review + create → create
5. After deployment, go to the resource.

Social Network Platform +

Overview
Boards
Repos
Pipelines
Pipelines
Environments
Library
Test Plans
Artifacts

#20260428.1 • Set up CI with Azure Pipelines

atech0718-oss/echoverse

Run now

This run is being retained as one of 3 recent runs by pipeline. [View retention leases](#)

Summary Code Coverage

Manually run by Mohammed Dhariq I B

Repository and version
atech0718-oss/echoverse
1st main • 11822c1d

Run started and elapsed
Today at 7:46 PM
0:20s

Related
0 work items
0 artifacts

[View change](#) [Parameters](#)

Tests and coverage
Get started

Stages Jobs

Stage	Jobs
Stage 1 job completed	7s
Job	3s

Rerun Stage

Steps to run Load Test.

1. Open the resource → click create Test
2. Enter test name and description
3. Provide the target API URL
4. Click create to run the test.

Pipelines:

Azure DevOps pipeline is used to automate build, testing, and deployment by connecting a Github project. It ensures every code commit is verified for reliability.

Steps:

1. connect Github to Azure DevOps and create a pipeline.
2. Add `azure-pipelines.yml` file in the repository.

3. YML code:

trigger:

```
- main # Trigger pipeline when changes are pushed to the main branch.
```

pool:

```
vmImage: ubuntu-latest # Use a hosted Ubuntu agent.
```

Steps:

```
# Step 1: Checkout the code from Github
```

```
- checkout: self
```

```
# Step 2: set up Node.js environment
```

```
- task: NodeTool@0
```

inputs:

```
versionSpec: '18.x'
```

displayName: 'Set up Node.js'

Step 3: Install dependencies

- script: |

npm install

displayName: 'Install dependencies'

Step 4: Run unit tests

- script: |

npm test

displayName: 'Run tests'

Step 5: Build the project

- script: |

npm run build

displayName: 'Build social network Platform'

Step 6: Publish build artifact

- task: PublishBuildArtifacts@1

inputs:

Path to Publish: 'dist'

Artifact Name: 'snp-build'

displayName: 'Publish Artifact'

4. Commit code to trigger the pipeline and monitor results.

Result:

Successfully performed load testing to analyse application performance and demonstrated an azure devops pipeline for automated build, testing and deployment.

Ex: no: 10

Date: 15/04/26

GitHub: Project structure & Naming conventions.

Aim:

To provide a clear and organised view of the social network platform project's folder structure and file naming conventions, helping contributors and users easily understand, navigate and extend the project.

GitHub Project structure:

The social network platform follows a modular, microservices - inspired folder structure for scalability and maintainability. Each service is separated into its own module with clear responsibilities.

Naming conventions:

Files and folders:

- Folders: camel case for service folders (e.g: user-service, post-service)
- controllers: Pascal case noun + 'Controller' (e.g. authController.js, postController.js)
- Model: Singular (User.js)
- Routes: Routes.js
- components / Pages: Pascal case (Postcard.jsx, LoginPage.js)
- Tests: .test.js

main 1 Branch 0 Tags

Go to file

Add file

Code

About

No description, website, or topics provided.

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Contributors 1

alitech0718-oss

Languages

TypeScript 93.6% PLpgSQL 5.0% Other 1.4%

Suggested workflows

Based on your tech stack



SLSA Generic generator

Configure

Generate SLSA provenance for your existing release workflows



Datadog Synthetics

Configure

Run Datadog Synthetic tests within your GitHub Actions workflow



Webpack

Configure

Build a Next.js project with rpm and webpack

alitech0718-oss	Set up CI with Azure Pipelines	1182261 · 3 hours ago	54 Commits
public	template: vite_react_shadon_ts_2026-03-20	2 years ago	
src	Limit views to unique per post	2 weeks ago	
supabase	Changes	2 weeks ago	
.env	Changes	3 weeks ago	
.gitignore	template: vite_react_shadon_ts_2026-03-20	2 years ago	
README.md	template: vite_react_shadon_ts_2026-03-20	2 years ago	
azure-pipelines.yml	Set up CI with Azure Pipelines	3 hours ago	
bun.lock	Work in progress	2 weeks ago	
bun.lockb	template: vite_react_shadon_ts_2026-03-20	2 years ago	
components.json	template: vite_react_shadon_ts_2026-03-20	2 years ago	
eslint.config.js	template: vite_react_shadon_ts_2026-03-20	2 years ago	
index.html	Reprint FKs to profiles	3 weeks ago	
package-lock.json	template: vite_react_shadon_ts_2026-03-20	2 years ago	
package.json	Connected Lovable Cloud	3 weeks ago	
playwright-fixture.ts	template: vite_react_shadon_ts_2026-03-20	2 years ago	
playwright.config.ts	template: vite_react_shadon_ts_2026-03-20	2 years ago	
postcss.config.js	template: vite_react_shadon_ts_2026-03-20	2 years ago	
tailwind.config.ts	Connected to Lovable Cloud	3 weeks ago	
tsconfig.app.json	Connected to Lovable Cloud	3 weeks ago	
tsconfig.json	Connected to Lovable Cloud	3 weeks ago	
tsconfig.node.json	template: vite_react_shadon_ts_2026-03-20	2 years ago	
vite.config.ts	template: vite_react_shadon_ts_2026-03-20	2 years ago	
vitest.config.ts	template: vite_react_shadon_ts_2026-03-20	2 years ago	

README

Branches:

- main - stable production code
- develop - integration branch for features
- feature / <feature-name> (e.g., feature/user-registration)
- bugfix / <bug-name> (e.g., bugfix/login-token-expiry)
- hotfix / <issue> (e.g., hotfix/moderation-crash)

commit messages:

- feat: add user registration endpoint
- fix: resolve duplicate friend request bug
- refactor: clean up moderation service logic
- test: add unit tests for post controller
- docs: update README with setup instructions.

Result:

The GitHub repository clearly displays the organised project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the social network platform codebase.